

Multi-Agent Scientific Paper Review

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File	Final_paper_kjns_210426 (1).pdf
Words	12,362
Sections Detected	preamble, introduction, methodology, experiments, discussion, section_vi, conclusions, section_vii, biogra
Review Mode	DEEP
Model	llama3.2

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Editorial Decision

MINOR REVISION

Weighted Score: 3.7 / 5.0
Confidence: 75%

Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

Criterion	Score	Rating	Visual
Originality & Novelty	4.0	Good	■■■■■
Manuscript Structure	4.0	Good	■■■■■
Methodology & Equations	4.0	Good	■■■■■
Statistical Analysis	3.5	Good	■■■■■
Results & Evidence	4.0	Good	■■■■■
Discussion & Conclusions	4.2	Good	■■■■■
Figures & Tables	4.0	Good	■■■■■
Scientific Writing	2.5	Weak	■■■■■
References & Citations	2.5	Weak	■■■■■
Ethics & Transparency	3.5	Good	■■■■■

Detailed Agent Reviews

TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: 4.0 / 5.0 | Confidence: 100%

Strengths:

- [TITLE] The title clearly signals the core method and application domain.
- [ABSTRACT] The abstract provides substantial methodological and validation detail.
- [ABSTRACT] Quantitative result signals are present in the abstract.
- [KEYWORDS] 8 keywords were detected for indexing and discoverability.

Major Comments:

- [TITLE] is informative and reflects the actual contribution.

Minor Comments:

- [ABSTRACT] could be more concise.
-

StructureReviewer

Manuscript structure and logical coherence

Score: 4.0 / 5.0 | Confidence: 80%

Strengths:

- The manuscript presents a clear research problem and explicitly declares the research question.
- The structure flow is generally logical, with a clear progression from introduction to conclusions.

Weaknesses:

- The section balance could be improved, as Section VI seems to repeat content from Section V.

Minor Comments:

- Consider adding more details about the preprocessing steps applied to the measured signals in Section IV.

Recommendations:

- Review and revise Section VI to ensure it adds new insights or perspectives not covered in Section V.
-

MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: 4.0 / 5.0 | Confidence: 80%

Strengths:

- Integrates physical admissibility constraints into the neural mapping
- Enables structural parameter estimation via deterministic forward inference

Major Comments:

- The sequence of sections could be improved for better logical flow
- More details on preprocessing steps and signal processing techniques would enhance the reproducibility of the results

Recommendations:

- Consider adding more context or background information on the EANN framework and its application in structural parameter identification
 - Provide more detailed explanations of the physical constraints used in the proposed method
-

StatisticsReviewer

Statistical methods, tests, and reporting quality

Score: 3.5 / 5.0 | Confidence: 60%

Strengths:

- Bootstrap confidence intervals and Holm correction are used correctly
- RMSE, MAE, R^2 , CV, and Pareto indicators are reported and interpreted appropriately
- The manuscript reports multiple statistical validation elements, including independent runs, nonparametric tests, correction procedures, confidence intervals, and performance/dispersion metrics.

Major Comments:

- The manuscript could benefit from a more detailed discussion on the assumptions of the Kruskal-Wallis test
- The use of non-parametric tests may be justified in this context, but explicit justification is not provided

Recommendations:

- Consider using more robust statistical methods for parametric identification
 - Provide a more detailed explanation of the assumptions underlying the Kruskal-Wallis test
-

FiguresTablesEquationsReviewer

Figures, tables, equations — quality and coherence with text

Score: 4.0 / 5.0 | Confidence: 62%

Strengths:

- 6 figure references were detected in the manuscript.
- 10 table references were detected in the manuscript.
- 52 sequentially numbered equations were detected, suggesting a substantial mathematical formulation.

Weaknesses:

- The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.

Minor Comments:

- The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.

Recommendations:

- Manually inspect captions, axes, units, table headers, and variable definitions in the final PDF.
-

ResultsReviewer

Results validity and empirical support

Score: 4.0 / 5.0 | Confidence: 80%

Strengths:

- The manuscript presents a clear research objective and framework for the problem.

Weaknesses:

- Some results, such as Pareto analysis and hypervolume/spread indicators, are not clearly explained or tied to figures/tables.
- Limitations of results are not explicitly acknowledged.

Major Comments:

- The manuscript could benefit from more detailed explanations of the Pareto analysis and hypervolume/spread indicators.
- It is unclear how the limitations of the proposed approach impact its robustness and significance.

Recommendations:

- Provide a clear explanation of the Pareto analysis and hypervolume/spread indicators, including their relevance to the problem and how they are used in the manuscript.
 - Acknowledge the limitations of the proposed approach and discuss potential avenues for improvement.
-

DiscussionConclusionsReviewer

Discussion depth and conclusions validity

Score: 4.2 / 5.0 | Confidence: 80%

Weaknesses:

- The discussion could benefit from a more in-depth comparison with prior work, including a detailed analysis of the strengths and limitations of each algorithm.

Major Comments:

- The Discussion section could be improved by explicitly stating the theoretical implications of the proposed EANN framework on structural health monitoring pipelines and adaptive control architectures.

Recommendations:

- Consider adding more details about the experimental setup, data preprocessing, and potential sources of error to strengthen the discussion.
-

WritingReviewer

Scientific writing quality, clarity, and style

Score: 2.5 / 5.0 | Confidence: 80%

Major Comments:

- The manuscript could benefit from a clearer introduction that sets the stage for the research.
- The experimental results are presented in a way that makes it difficult to understand the findings.

Recommendations:

- Revise the introduction to make it more concise and clear.
 - Combine similar sentences in Section V to improve flow.
-

ReferencesReviewer

References quality, recency, and relevance

Score: 2.5 / 5.0 | Confidence: 70%

Weaknesses:

- The references section could benefit from more recent publications (2022 and later) to reflect current state-of-the-art developments in the field.
- There is a lack of diversity in terms of research groups, countries, or institutions represented in the references.

Major Comments:

- While the manuscript provides a good overview of the current state of the art in magnetorheological dampers and system identification, it could benefit from more recent and diverse references to strengthen its arguments.
- The inclusion of seminal works is a strength, but the manuscript could further improve by providing more context or discussion on the limitations and potential biases of these foundational papers.

Minor Comments:

- The parser detected approximately 34 references; assess recency and relevance from the visible list rather than treating the section as empty.

Recommendations:

- Consider adding more recent conference papers (2022 and later) to the references section to reflect current advancements in the field.
 - Include a broader range of research groups, countries, or institutions to enhance the diversity of perspectives represented in the manuscript.
-

EthicsLimitationsReviewer

Research ethics, limitations, and transparency

Score: 3.5 / 5.0 | Confidence: 60%

Strengths:

- No clear human/animal-subject signal was detected in the parsed manuscript.

Weaknesses:

- Conflict-of-interest declaration was not detected.

Major Comments:

- Conflict-of-interest declaration was not detected.

Recommendations:

- Include a conflict-of-interest declaration.

Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

- [1] [TITLE] is informative and reflects the actual contribution.
- [2] The sequence of sections could be improved for better logical flow
- [3] More details on preprocessing steps and signal processing techniques would enhance the reproducibility of the results
- [4] The manuscript could benefit from a more detailed discussion on the assumptions of the Kruskal-Wallis test
- [5] The use of non-parametric tests may be justified in this context, but explicit justification is not provided
- [6] The manuscript could benefit from more detailed explanations of the Pareto analysis and hypervolume/spread indicators.
- [7] It is unclear how the limitations of the proposed approach impact its robustness and significance.
- [8] The Discussion section could be improved by explicitly stating the theoretical implications of the proposed EANN framework on structural health monitoring pipelines and adaptive control architectures.
- [9] The manuscript could benefit from a clearer introduction that sets the stage for the research.
- [10] The experimental results are presented in a way that makes it difficult to understand the findings.
- [11] While the manuscript provides a good overview of the current state of the art in magnetorheological dampers and system identification, it could benefit from more recent and diverse references to strengthen its arguments.
- [12] The inclusion of seminal works is a strength, but the manuscript could further improve by providing more context or discussion on the limitations and potential biases of these foundational papers.

Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

- [1] [ABSTRACT] could be more concise.
- [2] Consider adding more details about the preprocessing steps applied to the measured signals in Section IV.
- [3] The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.
- [4] The parser detected approximately 34 references; assess recency and relevance from the visible list rather than treating the section as empty.

Specific Improvement Recommendations

1. Review and revise Section VI to ensure it adds new insights or perspectives not covered in Section V.
2. Consider adding more context or background information on the EANN framework and its application in structural parameter identification
3. Provide more detailed explanations of the physical constraints used in the proposed method
4. Consider using more robust statistical methods for parametric identification
5. Provide a more detailed explanation of the assumptions underlying the Kruskal-Wallis test
6. Manually inspect captions, axes, units, table headers, and variable definitions in the final PDF.
7. Provide a clear explanation of the Pareto analysis and hypervolume/spread indicators, including their relevance to the problem and how they are used in the manuscript.
8. Acknowledge the limitations of the proposed approach and discuss potential avenues for improvement.
9. Consider adding more details about the experimental setup, data preprocessing, and potential sources of error to strengthen the discussion.
10. Revise the introduction to make it more concise and clear.
11. Combine similar sentences in Section V to improve flow.

12. Consider adding more recent conference papers (2022 and later) to the references section to reflect current advancements in the field.

This review was generated automatically by the Multi-Agent Paper Review System using Phi-3 Mini (Microsoft) running locally via Ollama. No manuscript content was transmitted to external servers. Rubrics based on Elsevier, IEEE, and Emerald Q1 peer review standards. This document does not constitute an official editorial decision.